

Cell: a rectangle within a spreadsheet where data can be positioned.

Formula: a mathematical calculation using +, -, x or.

Function: a ready-made formula representing a complex calculation.

Validation: the process of checking data matches acceptable rules.

Uses of Spreadsheets

It is used to manipulate numbers, perform calculations, present summary data and make predictions.

Spreadsheet Elements

Cells, Rows and Columns

A row consists of a horizontal set of cells. A column consists of a vertical set of cells. A cell is the intersection of a row and a column. A cell is referenced by the column letter and row number. A cell that is in column B and row 4 has a cell reference of B4.

Ranges

A range is a set of one or more cells. A range can be named so that the cell or cells can be referenced by the name instead of the cell reference. Range names are easier to remember and make reference to than cell references.

Worksheets

A worksheet is a set of rows and columns. It is possible to have more than one worksheet within a workbook. Worksheets can be used to separate information such as having a worksheet for sales in each country, a worksheet for sales for each year, a worksheet for exam results for each class.

Formulae

A formula uses basic arithmetic calculations. These are plus, minus, divide and multiply, for example =B5+B6. There is no limit to the number of arithmetic calculations that can be used, for example =(B5+B6) *3/100-(D1-D2).

	A	B	C	D	E
3	Job	Hours	Cost	Quantity	Total
4	A5 Leaflet	1	\$ 12.50	3	=C4*D4
5	A4 Leaflet	1.5	\$ 18.75	1	
6	A3 Leaflet	2.5	\$ 25.00	2	

When a formula with **relative cell references** is copied to another cell, the **reference** changes based on a **relative** position of rows and columns.

Absolute Cell References

Absolute referencing is used when you do not want a cell reference to change when other cells are filled in, or when replicating formulae.

Named Cells and Ranges

Named cells can be used as absolute cell references. When referring to a named cell, whenever it is replicated it will still point to the same named cell.

	A	B	C
1	Tax rate	0.15	
2			
3	Product	Cost	Tax
4	6 × 4 frame	12	=B4*TaxRate
5	5 × 7 frame	15	=B5*TaxRate
6	10 × 8 frame	18	=B6*TaxRate
7	12 × 8 frame	21	=B7*TaxRate

Functions

A function is a ready-made formula representing a complex calculation, for example =SUM (A5:B10) or =AVERAGE (cost).

Function	Purpose	Example
SUM	Calculates the total of values within a range.	=SUM(B3:E3)
AVERAGE	Calculates the average of values within a range.	=AVERAGE(B3:E3)
MINIMUM	Calculates the smallest value within a range.	=MIN(B3:E3)
MAXIMUM	Calculates the largest value within a range.	=MAX(B3:E3)

Rounding Functions

Function	Purpose	Example
INTEGER	Returns the whole number value of a decimal number (the value before the decimal point).	=INT(A2)
ROUND	Rounds a number to the nearest decimal place specified. This example rounds the value in B2 to three decimal places.	=ROUND(A2,3)
ROUNDUP	Rounds a number up to the nearest decimal place specified. This example rounds up the value in B2 to two decimal places.	=ROUNDUP(A2,2)
ROUNDDOWN	Rounds a number down to the nearest decimal place specified. This example rounds down the value in B2 to four decimal places.	=ROUNDDOWN(A2,4)
MEDIAN	It is used to find the middle value when all data items are listed in ascending order.	=MEDIAN(\$C\$8:\$C\$80)
MODE	This function is used to find the piece of data (in this case test score) that occurs most frequently.	=MODE(\$C\$8:\$C\$80),

Example

	A	B	C	D	E
1	Original number	Integer	Round	Round up	Round down
2	25817.32817	=INT(A2)	=ROUND(A2,3)	=ROUNDUP(A2,2)	=ROUNDDOWN(A2,4)
3	3852.876985	=INT(A3)	=ROUND(A3,3)	=ROUNDUP(A3,2)	=ROUNDDOWN(A3,4)
4	928.2341	=INT(A4)	=ROUND(A4,3)	=ROUNDUP(A4,2)	=ROUNDDOWN(A4,4)
5	0.03256	=INT(A5)	=ROUND(A5,3)	=ROUNDUP(A5,2)	=ROUNDDOWN(A5,4)

Indices

If you need to calculate indices (powers), then the POWER function can be used. It uses the syntax =POWER (number, power). For example, =POWER (6,3) would raise 6 to the power of 3. It is also possible to use the ^ symbol and so =6^3 would also raise 6 to the power 3.

	A	B	C	D
1	Original number	Power 2	Power 3	Power 4
2	4	=POWER(A2,2)	=A2^3	=POWER(A2,4)
3	3	=POWER(A3,2)	=A3^3	=POWER(A3,4)
4	7	=POWER(A4,2)	=A4^3	=POWER(A4,4)

Date and Time Functions:

Function	Purpose	Example
DAY	Calculates the day part of a date.	=DAY(B1)
MONTH	Calculates the month part of a date.	=MONTH(B1)
YEAR	Calculates the year part of a date.	=YEAR(B1)
DATE	Calculates the date from a given year, month and day.	=DATE(B4,B3,B2)
HOUR	Calculates the hours part of a time.	=HOUR(B8)
MINUTE	Calculates the minutes part of a time.	=MINUTE(B8)
SECOND	Calculates the seconds part of a time.	=SECOND(B8)
TIME	Calculates the time from given hours, minutes and seconds.	=TIME(B9,B10,B11)
NOW	Gives the current date and time (can be formatted for just date or just time).	=NOW()

Text Functions

Function	Purpose	Example
CONCATENATE	Joins together text values.	=CONCATENATE =(A1,B1,C1) =A1&B1&C1 =A1& & B1&"@" &C1 & ".it.com"
LEFT	Extracts the furthest left characters.	=LEFT(A1,4)
Right	Extracts the furthest right characters.	=RIGHT(A1,2)
LEN	Calculates the length of a string.	=LEN(A1)
UPPER	Converts text into uppercase.	=UPPER(A1)
LOWER	Converts text into lowercase.	=LOWER(A1)

VLOOKUP

It is used to look for a value in a column and return a value from the same row.

=VLOOKUP(Search Value, Lookup Table, Column, Match)

=VLOOKUP(B3,Client.csv!\$A\$2:\$B\$8,2,FALSE)

Search Value is the value being looked up in the table. Lookup Table is the range of cells that define the location of the table. Column is the number of the column that should be returned as the result in the matching row of the value that was found. Match defines whether an exact match should be found - TRUE for approximate match or FALSE for exact match.

The HLOOKUP function uses the syntax:

=HLOOKUP(Search Value, Lookup Table, Row, Match)

=HLOOKUP(B6,\$B\$2:\$H\$3,2)

Search Value, Lookup Table and Match are the same as in the VLOOKUP function. Row is the number of the row that should be returned as the result in the matching column of the value that was found.

INDEX Function

The INDEX function returns values from a given location in a table, where the user specifies the row and column position of the item in the table.

=INDEX(\$A\$16:\$C\$40,2,2), which looks in the range of cells between A16 and C40, finds the second row in that range, then returns the value held in the second column.

=INDEX(\$A\$16:\$C\$40,3,2) finds the third row in that range and returns the second cell within that row.

MATCH Function

The MATCH function searches for a specified item in a range of cells and returns the relative position of that item in the range.

=MATCH(\$B\$2,\$B\$10:\$B\$165,0), which matches the contents of cell B2 (in this case the test data 30) with the first 30 it finds within the range B10 to B165. The final '0' instructs *Excel* to find an exact match.

Conditional Functions

The functions you have seen so far will always perform their calculation. However, a conditional function will only perform the calculation if certain criteria are met.

IF Statement

=IF(A1=5,A2*0.05,"No discount")

=IF(C3>=10,"Very experienced","Not experienced")

Nested IF

It is possible to have an IF function nested within another IF function. This can be useful if there are two or more alternative outputs. Sometimes the same result can be achieved with a lookup function.

=IF(C3>=10,"Very experienced",IF(C3>=5,"Experienced","Not experienced"))

Function	What it does
=IF(B6="C","Car","Not a car")	Checks the contents of cell B6. If the contents match the letter "C" with no additional spaces, etc., then cell G1 displays "Car". If cell B6 contains anything other than the letter "C", then cell G1 displays "Not a car".
=IF(B7=3,"March","Not March")	Checks the contents of cell B7. If the contents are the value 3, then cell G1 displays "March". If cell B7 contains anything other than the value 3, then cell G1 displays "Not March".
=IF(B7<4,"1st Quarter","")	Checks the contents of cell B7. If the contents are less than the value 4, then cell G1 displays "1st Quarter". If cell B7 contains a value of 4 or more, then cell G1 displays a null string (no contents are shown).
=IF(B7<=3,"1st Quarter","")	Checks the contents of cell B7. If the contents are less than or equal to 3, then cell G1 displays "1st Quarter"; if not, cell G1 displays a null string.
=IF(OR(B7=4,B7=5,B7=6),"2nd Quarter","")	Checks the contents of cell B7. If the contents are any of the values 4, 5 or 6, cell G1 displays "2 nd Quarter"; if not, cell G1 displays a null string.

=IF(AND(B7>3,B7<7),"2nd Quarter", "")	Checks the contents of cell B7. If the contents are greater than 3 AND less than 7, cell G1 displays "2nd Quarter"; if not, cell G1 displays a null string.
=IF(NOT(OR(B7<4,B7>6)),"2ndQuarter", "")	Checks the contents of cell B7. If the contents are NOT less than 4 OR NOT greater than 6, cell G1 displays "2nd Quarter"; if not, cell G1 displays a null string.

IFS Function

=IFS(B2="Alpha","London",B2="Beta","Milan",B2="Gamma","Tunis")

The **IFS** function looks at a number of given conditions and performs a different operation if one of these conditions is met. If none of the conditions are met it returns an error message, so careful use of error trapping may be needed with this function. This is just a more efficient method of nesting IF statements.

ISERROR Function

It is used to detect if an error will occur when a formula is used, and returns True if there is an error or False otherwise. Although this can be used to detect errors, there is a better alternative that detects the error and allows us to trap the error so that it does not appear as *Excel* would generate it. The IFERROR function can be used to indicate the error or in some cases correct the error.

=IFERROR(IFS(B2="Alpha","London",B2="Beta","Milan",B2="Gamma","Tunis"),"Project location unknown")

Conditional Statistical Functions

Function	Example
COUNTA	=COUNTA (H3:H200)

COUNTIF	=COUNTIF (H3:H200,"Yes") Only counts cells that contain "Yes".
COUNTIFS	=COUNTIFS(B5:B18,"None",C5:C18,"None",D5:D18,"None") It will check the first cell in the range B5 to B18 and if it contains the string "None", will move on to check the cell in the same row within the range C5 to C18. If this cell also contains the string "None", it will then check in the same row within the range D5 to D18 and will only count if it also contains the string "None", before moving on to test the next row. It does this until it reaches cell D18.
AVERAGEIF	=AVERAGEIF (J3:J200,">0") Only finds the average of positive numbers.
SUMIF	=SUMIF (J3:J200,"<0") Only adds up negative numbers. =SUMIF(\$D\$24:\$D\$52,A9,\$F\$24:\$F\$52) which compares the contents of each cell in the range D24 to D52 with the contents of cell A9. If these values or strings match, then the value contained in the corresponding cell in the row within the range F24 to F52 is added to the total.
SUMIFS	=SUMIFS(\$F\$24:\$F\$52,\$D\$24:\$D\$52,A14,\$E\$24:\$E\$52,"Full") . This adds the value of each row held within the range F24 to F52 to the total, only if both these criteria are met: » the contents of the cell (in this row) in the range D24 to D52 contain the same as cell A14 » the contents of the cell (in this row) in the range E24 to E52 contain the label "Full".

Validation Rules

The validation rules that can be used in spreadsheet software, include lookup in list, range, type (integer and real) and length.

A range check can be set up with the following properties:

- between two values
- equal to a value
- not equal to a value
- greater than a value
- greater than or equal to a value
- less than a value
- less than or equal to a value.

Validation Applied to a Spreadsheet

Type of test data	Purpose
Valid (normal/acceptable data)	Data that should pass the validation rule.
Invalid (erroneous/ unacceptable data)	Data that should generate an error message.
Extreme valid	Data that will only just pass the validation rule because it is on the boundary of acceptable data.
Extreme invalid	Data that will only just generate an error message because it is on the boundary of unacceptable data.

EXAMPLE

Here is some test data that could be used to test the validation rule whole number BETWEEN 5 and 20:

Test data input value	Type of test data	Expected result
10	Valid	Accepted
5	Extreme valid	Accepted
20	Extreme valid	Accepted
3	Invalid	Error message
-10	Invalid	Error message
abc	Invalid	Error message
10.5	Invalid	Error message
4	Extreme invalid	Error message
21	Extreme invalid	Error message

Effectiveness of Test Plans for a Spreadsheet Model

Testing of validation rules is just one aspect of testing. All formulae and functions also need testing. To test a formula or function, it is necessary to identify input data that will be used for the calculation and the expected output data (result of calculation). The expected output data should be calculated using traditional methods and then compared with the result from the spreadsheet.

Other aspects such as cell and worksheet protection and conditional formatting can also be tested. It is not possible to test every possible input value, but a good range should be used, especially extreme values which are small and large. Invalid input values should only be used if validation rules have been set up.

Verify Data Entry

When data that has been validated is not necessarily correct. It is therefore necessary to verify data. When inputting data into a spreadsheet, you should verify that the data has been input correctly by comparing what has been entered with the original source.

Filter Data

When there is a lot of data listed in a spreadsheet, users may want to view just some of that data. This can be achieved by filtering the data that is required. The term filter is used to relate to the user's mental model. When coffee is filtered, the granules are left in the filter while the flavored drink flows through the filter.

Filter	Purpose
Contains	Selects data in a column that includes the text specified.
Starts with	Selects data in a column that starts with the text specified.
Ends with	Selects data in a column that ends with the text specified.
Greater than	Selects data in a column that is greater than the value specified.
Less than	Selects data in a column that is less than the value specified.
Greater than or equal to	Selects data in a column that is greater than or equal to the value specified.
Less than or equal to	Selects data in a column that is less than or equal to the value specified.

Sort Data

It is possible to sort data into ascending or descending order. If only a selection of columns is sorted, then only the data within those columns will be sorted. To keep all data in each row together, do not highlight the data. To sort on more than one column, it is necessary to specify the sort order for each column and which column should be sorted first. If you wanted to sort by surname and then sort any matching surnames by forename, then you would sort by surname first. It's not just text and numbers that can be sorted into order. Dates and times can also be sorted.

Import and Export Data

Data can be imported from other formats such as a comma separated values (CSV) file, text (TXT) file or from a database. When importing, you should already have the spreadsheet software open and open the file to be imported from within the spreadsheet software. You will often have to ensure that when opening you have selected the option to show all files.

Data can also be exported so that it can be used in other software. If it is exported in a common format such as CSV or TXT, then other software will be able to recognize the rows and columns. Formats such as portable document format (PDF) will enable the data to be viewed by a wide variety of users but not manipulated, so PDF is good for showing output data.

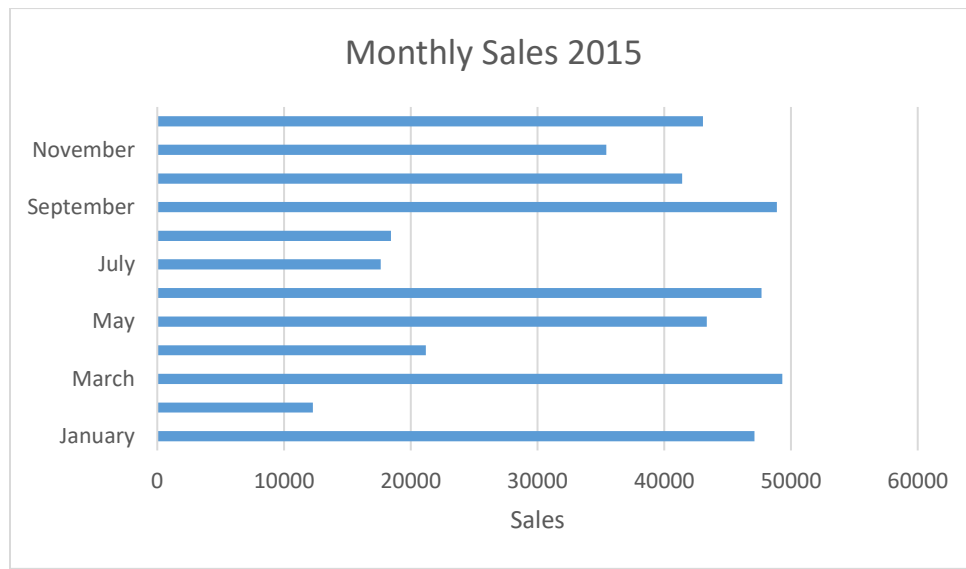
Graphs and Charts

Graphs or charts are used to show information in a way that makes it easy to compare values. This could be as simple as showing the monthly sales in a column or bar chart or it could be more complex and show how hospital waiting times vary during a month compared with their targets. Charts that show more than one set of data are known as comparative charts because data for one set can be compared against data for another.

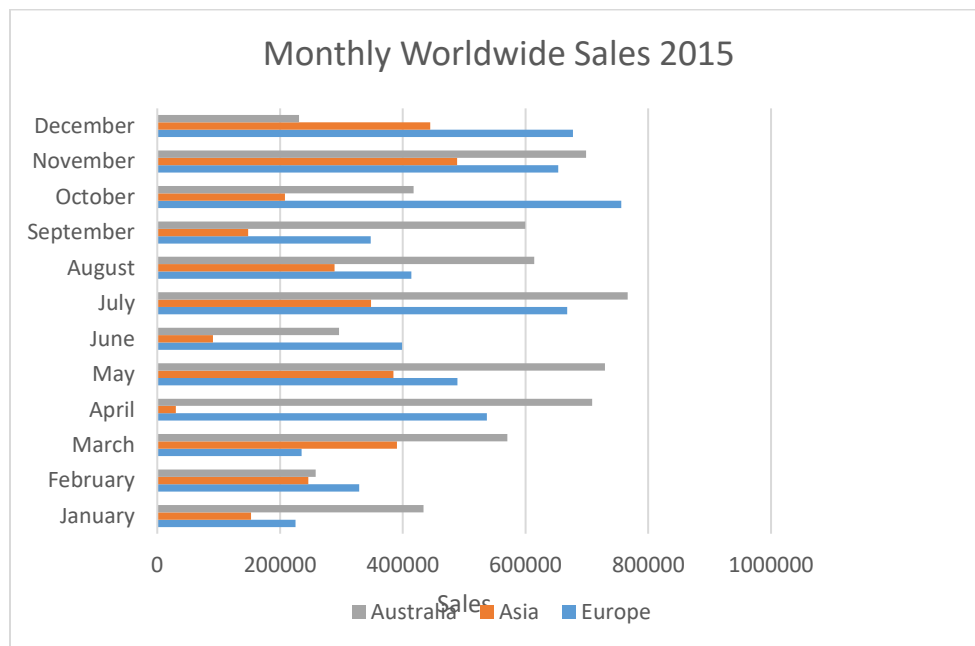
Graphs and charts include some examples of graphs and charts. The worksheet **Monthly Sales**

contains a bar chart showing the Monthly sales over a period of 12 months.

Notice how a chart title and x-axis title have been included:

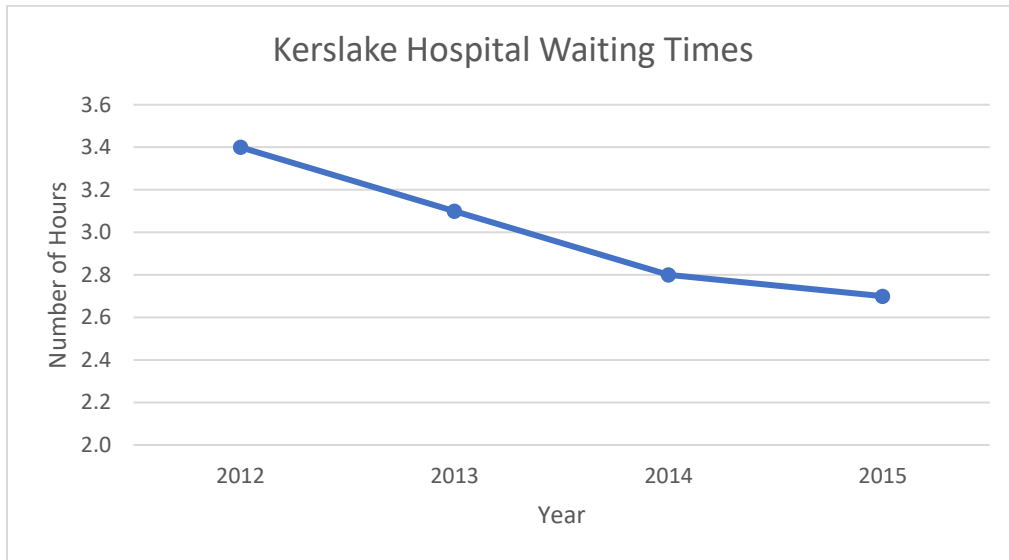


The comparative bar chart on the worksheet Worldwide Sales compares the sales of different continents for each of the months in 2015. Notice how a legend is used to show which continent is represented by which color:

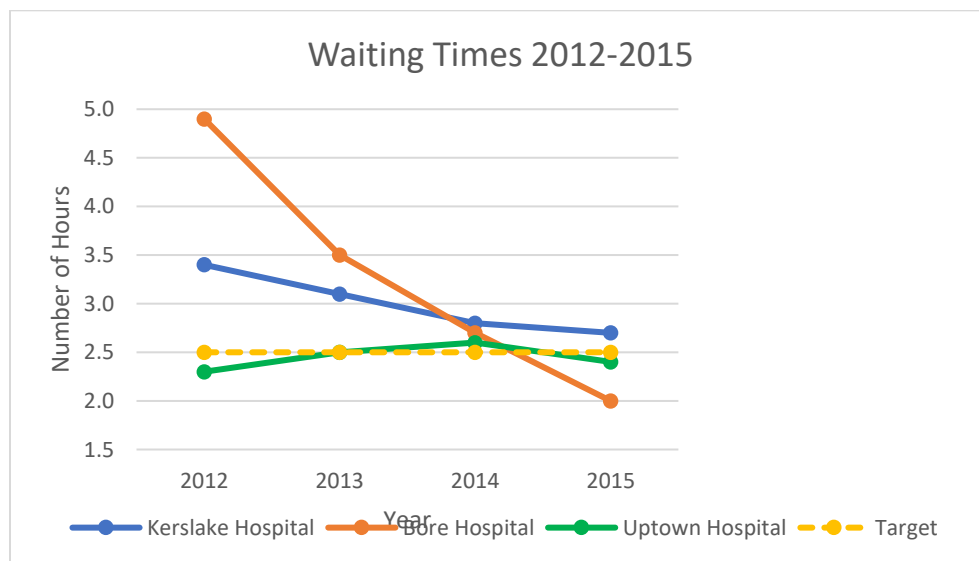


Line graphs are similar to bar charts in that they show the size of data, but they are mainly used to show how data changes over a period of time.

The first line graph on the worksheet Waiting List shows how waiting times at the Kerslake Hospital have fallen over a four-year period. Notice how the minimum value on for the x-axis scale is 2.0 hours and the maximum value for the y-axis scale is 3.6 hours. The y-axis scale has intervals of 0.2 hours. Both y and x-axes have been given axis titles.

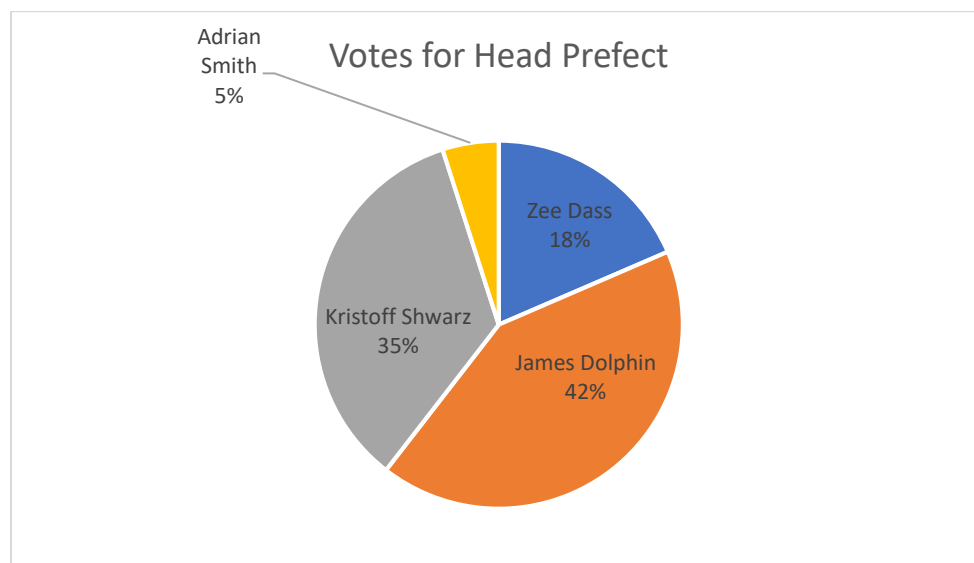


The comparative line graph on the worksheet Waiting List compares the waiting times of three hospitals over a four-year period with the target waiting time of 2.5 hours. Notice how the y-axis scale has intervals of 0.5 hours, the maximum value for the y-axis scale is 5.0 and the minimum value for the y-axis scale is 1.5.



Pie charts are used to show proportional data. This differs from bar charts or line graphs because instead of showing the actual values, the proportion or percentage of each data item is shown.

Graphs and charts contain a pie chart showing the proportion of votes received for each student standing in an election for head prefect of a school. Notice how a legend is not necessary because each legend has been labelled with the name of the data item. Each segment has also been labelled with the percentage of the total vote that each data item represents:



Type of graph / chart	Purpose
Bar/ column chart	Used to show a single series of data in columns or rows.
Comparative bar/ column chart	Used to compare more than one series of data in columns or rows.
Line graph	Used to show how a single series of data changes over a period of time.
Comparative line graph	Used to compare how more than one series of data changes over a period of time.
Pie chart	Used to show the proportion that is used by each item of data.

Create a Graph or Chart

When creating a graph or chart, you should ensure that it is labelled appropriately.

Label	Purpose
Title	A brief overview of what the graph or chart is showing.
Legend	A key to show what the coolers used represent.
Value axis	Each of the axes (usually x and y) should include the values including any units.
Category axis	Each of the axes should include a title to state what the data represents.
Percentages	When using a pie chart, the percentages may need to be displayed for each segment.
Segment label	Instead of a legend, each segment in a pie chart could be labelled with its description.
Segment value	As well as, or instead of the percentage, each segment in a pie chart could be labelled with its quantity.
Scale	The interval between each value on an axis.
Axis scale maximum	The maximum value to be used on an axis.
Axis scale minimum	The minimum value to be used on an axis.

Pivot Tables (Class Discussion)

A pivot table can analyze and summaries data into a two-dimensional table. It is used to see patterns and trends in data.

Automate Operations in Excel (Class Discussion)